

Qatar-1b

R-1984

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_210615 · created 2026-08-07T01:08:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459380.8742 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.072 +/- 0.042
Transit depth (Rp/Rs)^2	0.51 +/- 0.6 [%]
Orbital Inclination (inc)	85.21 +/- 3.08
Airmass coefficient 1 (a1)	0.941 +/- 0.018
Airmass coefficient 2 (a2)	0.054 +/- 0.056
Scatter in the residuals of the lightcurve fit is	1.89 %
Best Comparison Star	#2 - [333, 70]
Optimal Aperture	2.93
Optimal Annulus	8.71
Transit Duration (day)	0.063 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.072 ± 0.042 vs 0.1463 (deviation 1.67σ)
Duration fit vs published	0.063 d vs 0.0692 d (deviation 0.31σ)
Mid-time O–C	-4.5 min
Depth SNR	1.6
Residual scatter	1.89 %

Light curve

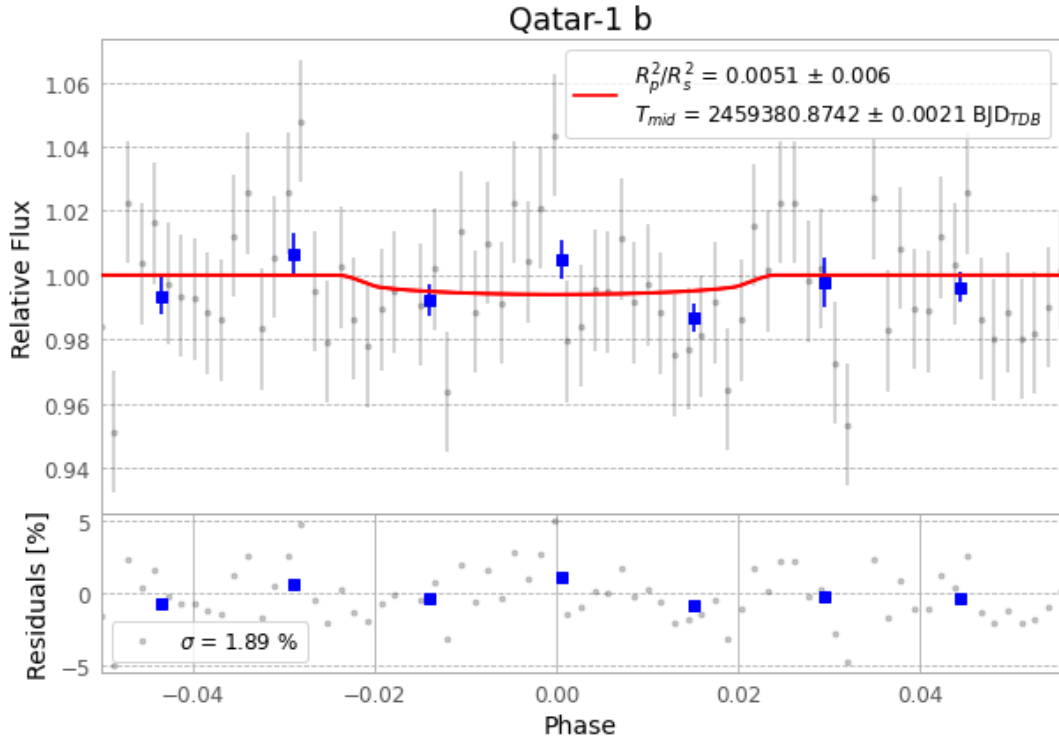


Figure 1 — FinalLightCurve_Qatar-1 b_2021-06-15.png (SHA-256 baf3f159d3bbafb9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [335, 263], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ac61fce5d97f26c0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

73 FITS frames (48 MB), Qatar-1210615071512.FITS → Qatar-1210615105112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-210615.log (b885d35fa9ec...), FinalLightCurve_Qatar-1 b_2021-06-15.png (baf3f159d3bb...), FinalLightCurve_Qatar-1 b_2021-06-15.csv (fc94167b48cf...)

Compute resources

Wall time 107.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 01:08Z.